NXT WAVE

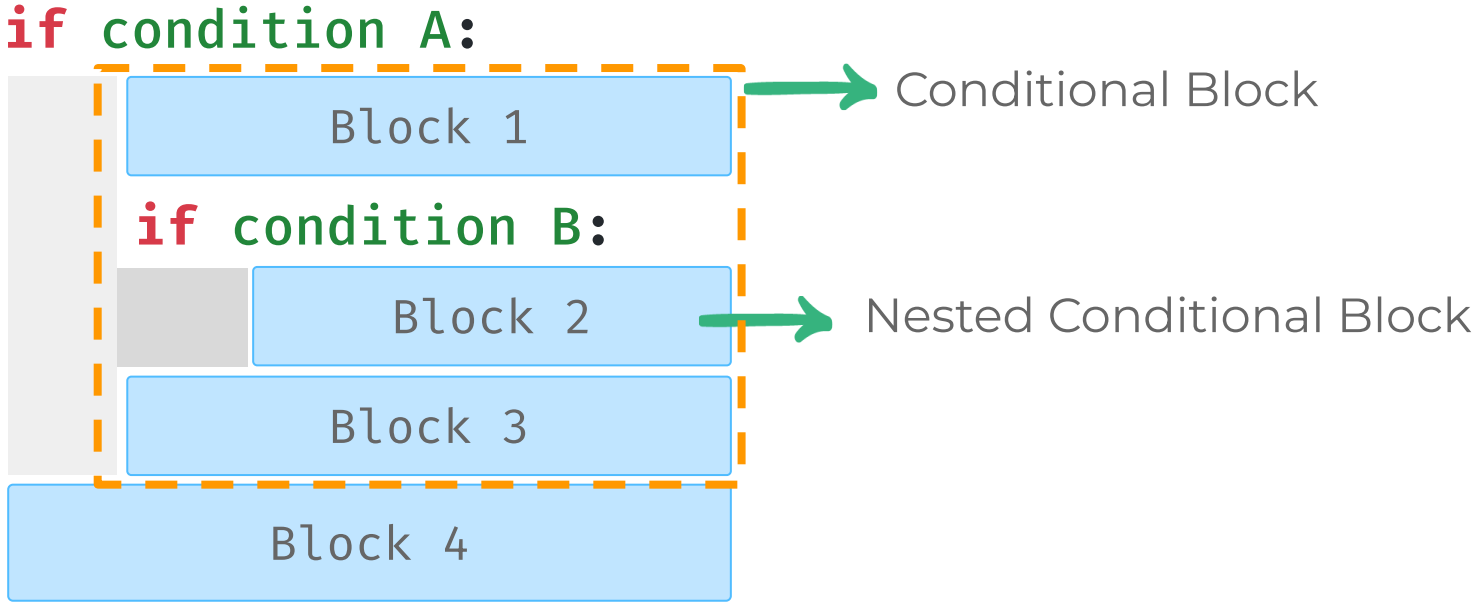
WELCOME

CCBP 4.0

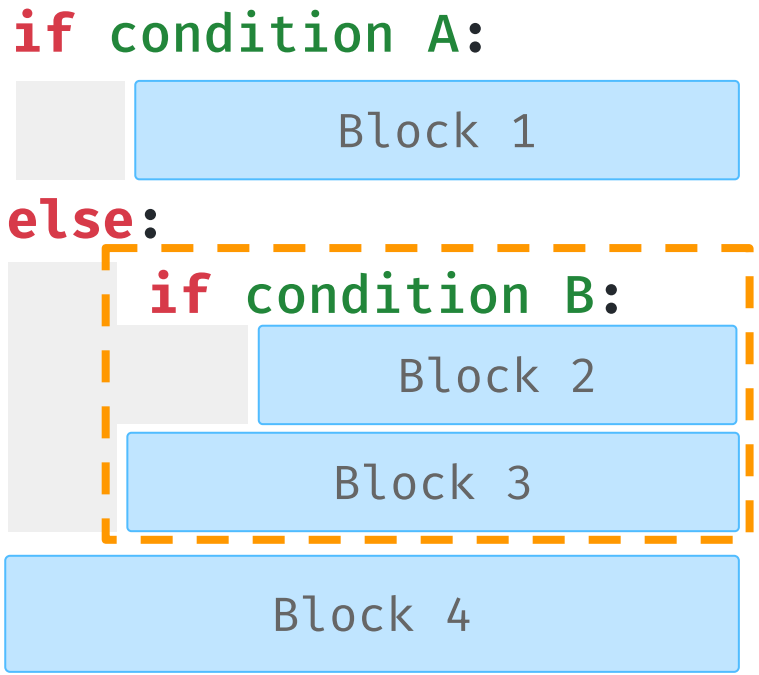
21st Century Skills

Loops

Nested Conditions



Nested Conditions



We can also write nested conditions in Else block

Recap

Nested Conditions

```
if condition A:
    Block 1
elif condition B:
    Block 2
else:
    Block 3
```

Nested Conditions

```
if condition A:
    Block 1
elif condition B:
    Block 2
elif condition C:
    Block 3
else:
    Block 4
```

We can add any number of `elif` blocks after `if` block

Recap

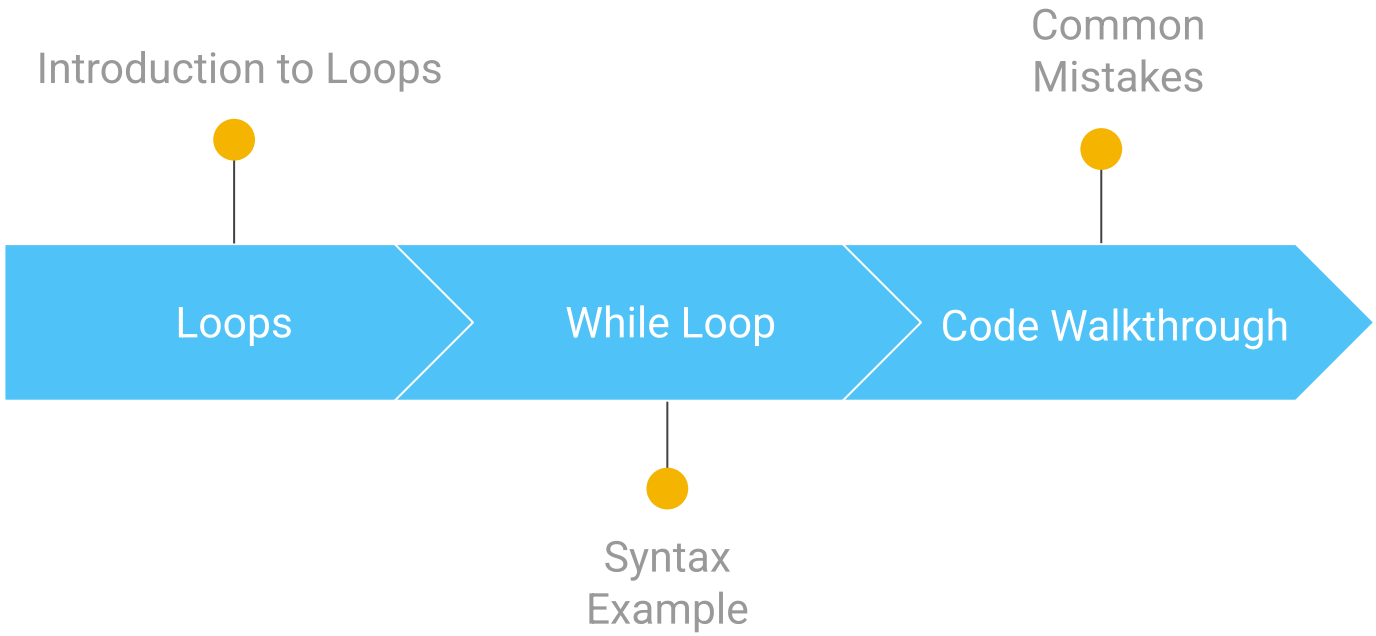
Nested Conditions

```
if condition A:
        Block 1
elif condition B:
        Block 2
elif condition C:
        Block 3
```

`else` block is not compulsory after `if` - `elif` blocks

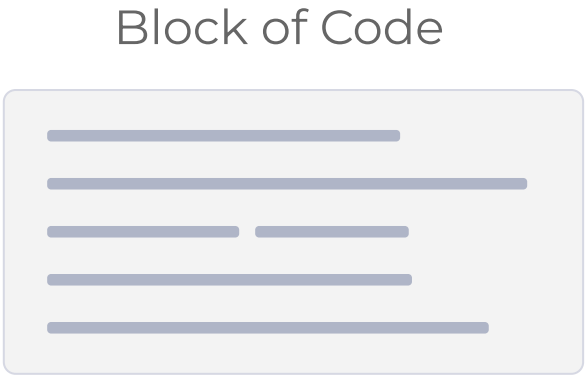
Agenda for Today's Session

Loops



Sequence of Instructions

Python executes code in a sequence
and each block of code
is executed once



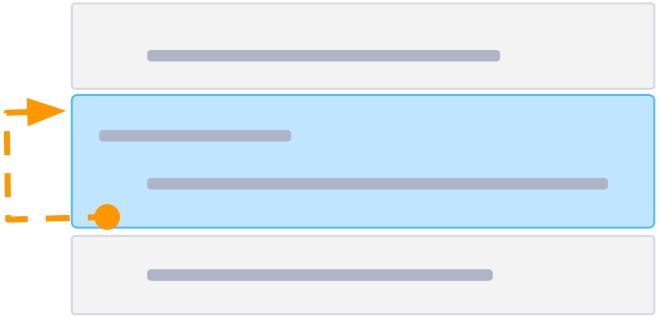
Question

Can we control the flow of execution to make a block of code to execute more than once?



Loops

Loops allow us to **execute**
a block of code **several times**



Loops

In Python,
there are two primary ways
for looping



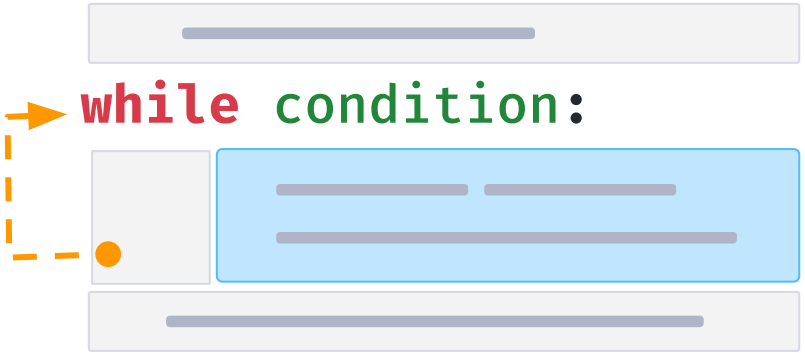
While Loop



For Loop

While Loop

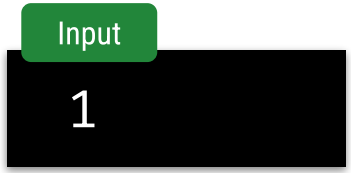
Used to execute a block of code several times as long as the condition is True

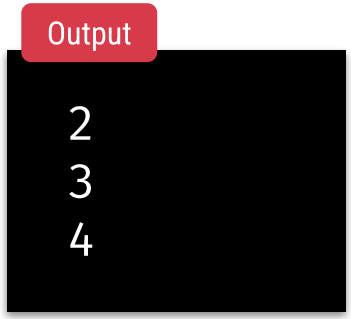


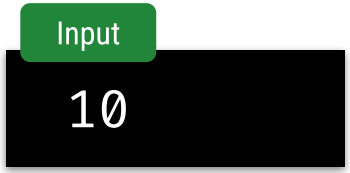
While Loop

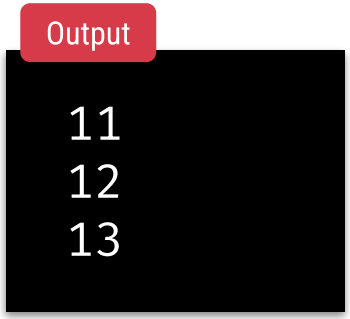
Example

Let’s write a code to print the next 3 consecutive numbers of the given integer input









Code

Code

```
a = int(input())
a = a + 1
print(a)
a = a + 1
print(a)
a = a + 1
print(a)
```

Input

```
5
```

Output

```
6
7
8
```

Code

Code

```
a = int(input())
a = a + 1
print(a)
a = a + 1
print(a)
a = a + 1
print(a)
```



We have to make
this block of code
execute 3 times,
instead of writing it 3 times

While Loop

```
while condition:
    Repeating
    Block of Code
```

Code

```
a = int(input())
while condition :
    a = a + 1
    print(a)
```



While Loop

Code

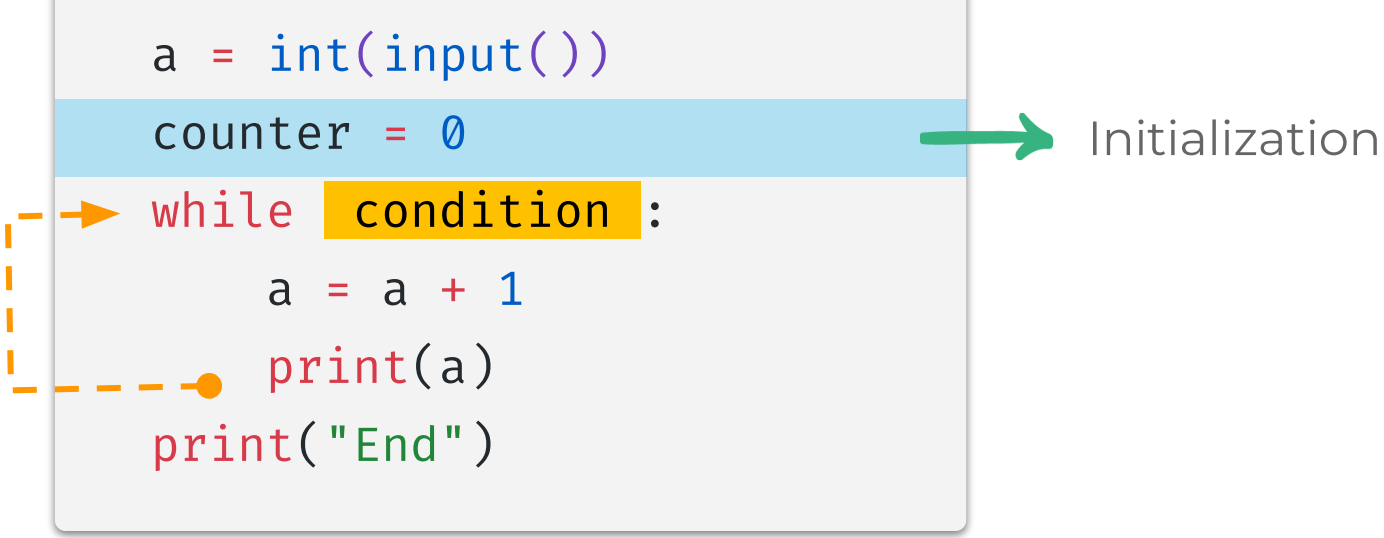
```
a = int(input())
while condition :
    a = a + 1
    print(a)
```

Let's initialize a counter variable to check if the loop has executed 3 times

While Loop

Code

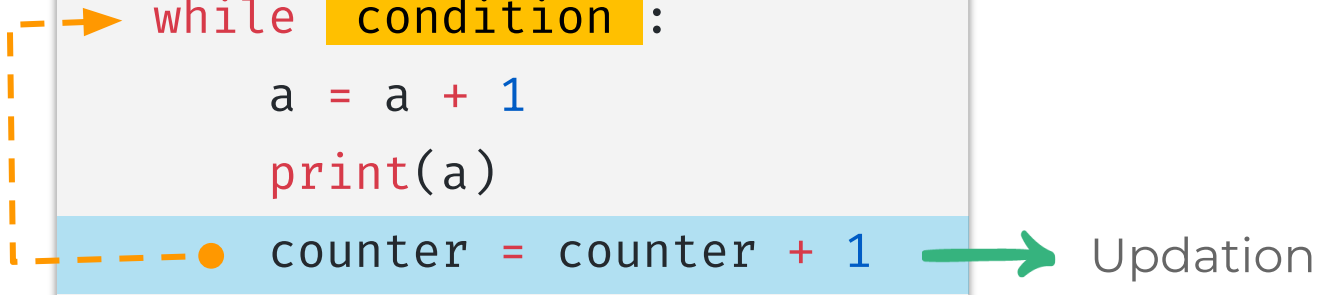
```
a = int(input())
counter = 0
while condition :
    a = a + 1
    print(a)
print("End")
```



While Loop

Code

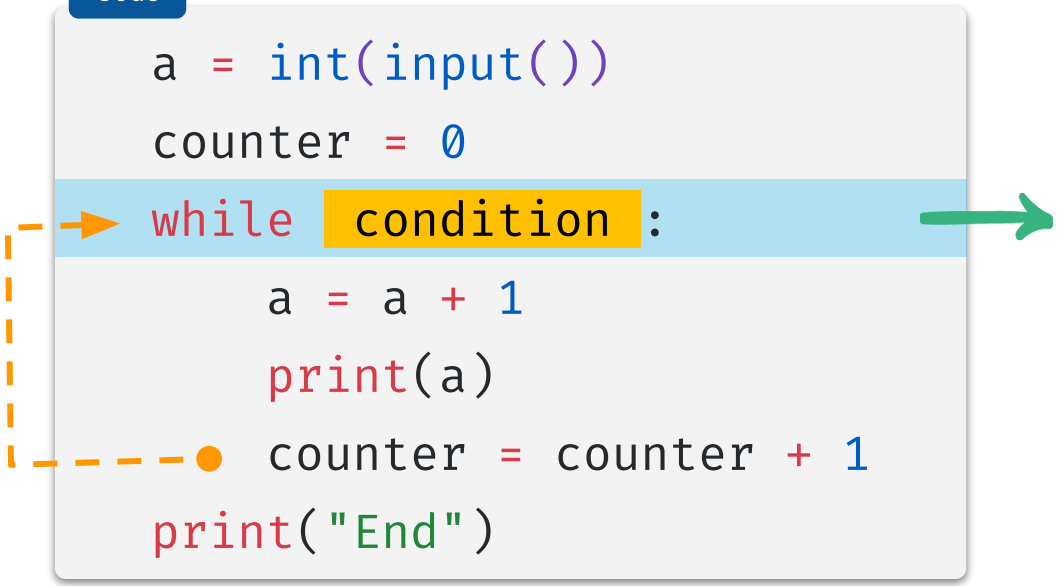
```
a = int(input())
counter = 0
while condition :
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



While Loop

Code

```
a = int(input())
counter = 0
while condition :
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

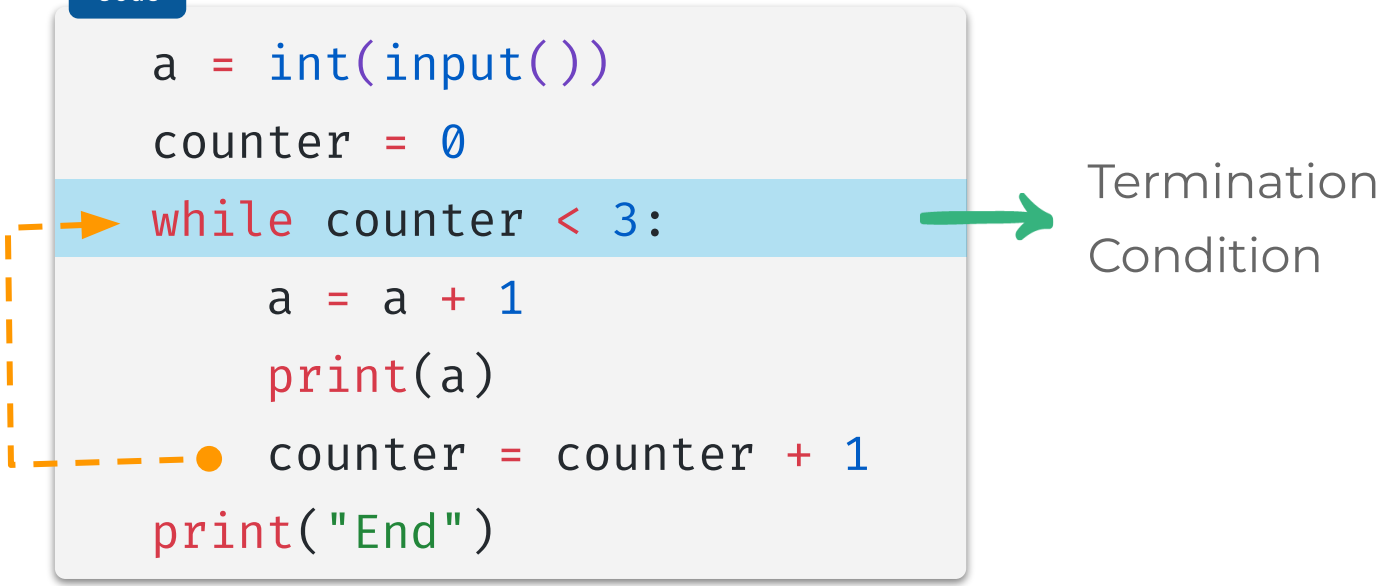


What should the condition be?

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



While Loop

Code

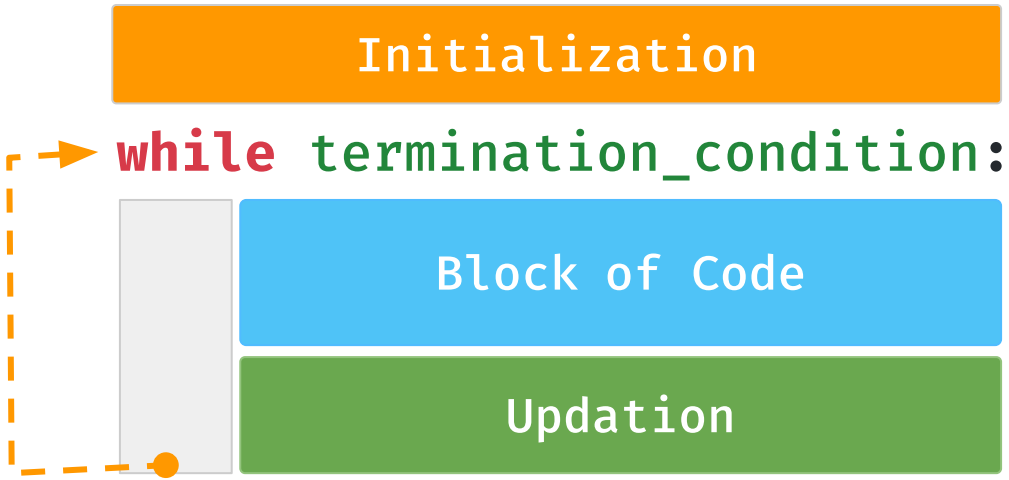
```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

Initialization

Termination Condition

Updation

While Loop


Initialization
while termination_condition:
Block of Code
Updation

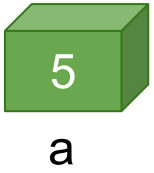
While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

Input

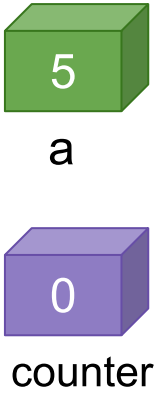
5



While Loop

Code

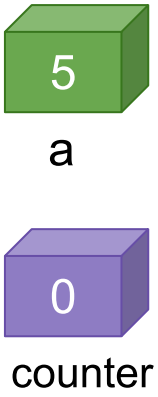
```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



While Loop

Code

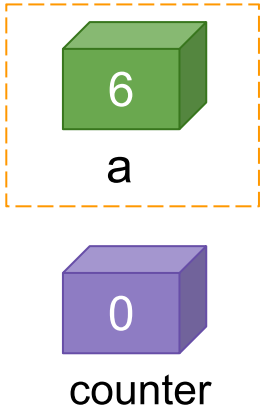
```
a = int(input())
counter = 0
while counter < 3:          True
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



While Loop

Code

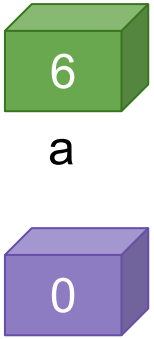
```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



Output

```
6
```

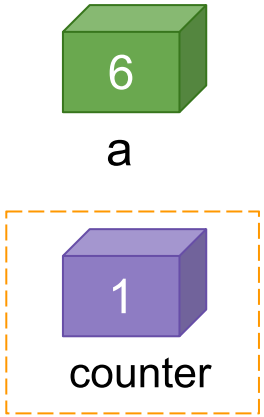
While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

Output

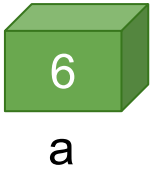
```
6
```



While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:          True
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



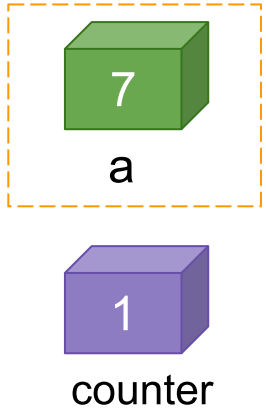
Output

```
6
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



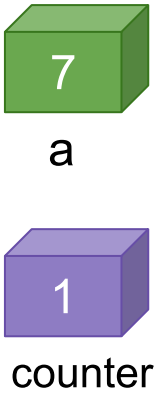
Output

```
6
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



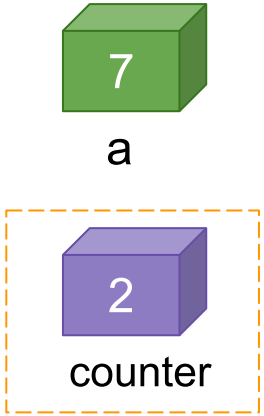
Output

```
6
7
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



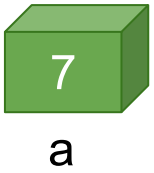
Output

```
6
7
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:          True
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



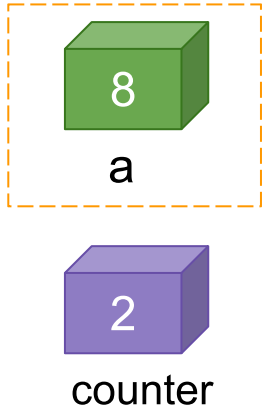
Output

```
6
7
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



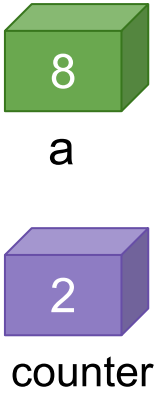
Output

```
6
7
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



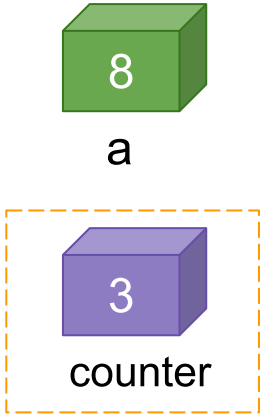
Output

```
6
7
8
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



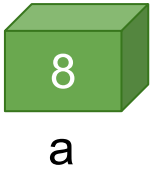
Output

```
6
7
8
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:          False
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```



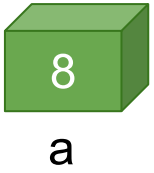
Output

```
6
7
8
```

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

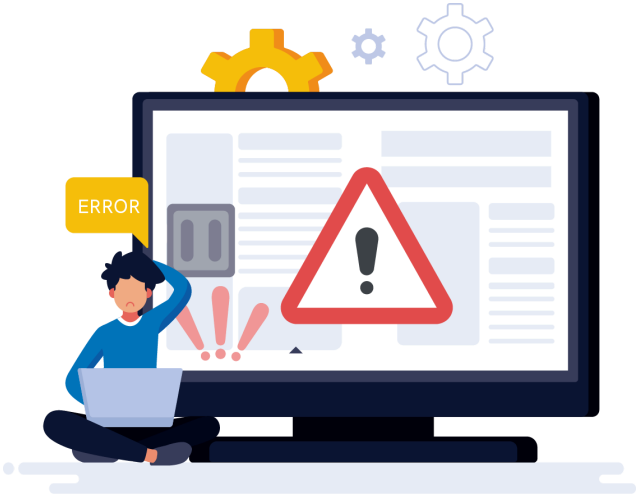


Output

```
6
7
8
End
```

Common Mistakes & Errors

Let's look into
common mistakes and errors
while using loops



Missing Initialization

Code

```
a = int(input())
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

Input

```
5
```

Output

```
NameError:
name 'counter'
is not defined
```

Incorrect Termination Condition

Code

```
a = int(input())
counter = 0
condition = (counter < 3)
while condition:
    a = a + 1
    print(a)
    counter = counter + 1
```

Input

```
10
```

What will the output be?

Incorrect Termination Condition

```
a = int(input())
counter = 0
condition = (counter < 3)
while condition:
    a = a + 1
    print(a)
    counter = counter + 1
```

Infinite Loop

while block will keep repeating as the value in `condition` variable is True

Incorrect Termination Condition

Code

```
a = int(input())
counter = 1
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

Input

```
1
```

Output

```
2
3
End
```

Not Updating Counter Variable

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
print("End")
```

Infinite Loop

As the value of **`counter`** is not updating

Question

Square Pattern

Write a program to print stars in N rows and N columns, where integer N is given as an input

Input

```
4
```

Output

```
* * * *
* * * *
* * * *
* * * *
```

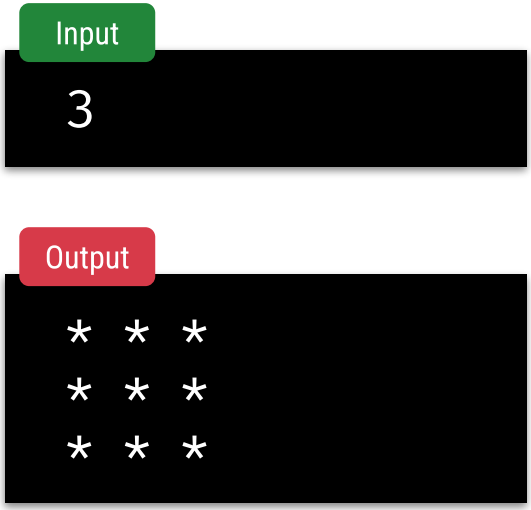
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Question

Input / Output Format

Input Format

First line will contain a positive integer



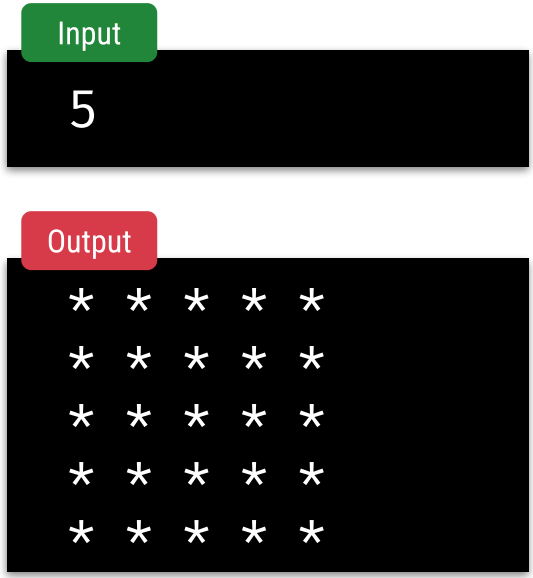
Question

Input / Output Format

Output Format

Print stars in N rows and N columns

*Note: There is a space after every **



Question

While Loop

Code

```
n = int(input())
counter = 0
while counter < n:
    print("* " * n)
    counter = counter + 1
```

Input

```
4
```

Output

```
* * * *
* * * *
* * * *
* * * *
```

NXT WAVE

Key Takeaways

- Loops
- While Loop
  - Syntax
  - Initialization
  - Termination Condition
  - Updation
- Common Mistakes and Errors
- Code Walkthrough



Practice



Daily Challenge

Identify The Mistake

Code

```
a = input()
counter = 0
length_of_a = len(a)
while counter < (length_of_a - 1):
    print(a[counter])
    counter = counter + 1
```

Input

```
Python
```

Expected Output

```
P
y
t
h
o
n
```

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Next Session

For Loop

Feedback



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THANK YOU

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ALL THE BEST